

4.25

(a)

```
data = read.csv("collegetown.csv")

data <- data %>%
  mutate(ln_price = log(price))

model <- lm(ln_price ~ sqft, data = data)

summary(model)
```

```
##
## Call:
## lm(formula = ln_price ~ sqft, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.32528 -0.19199  0.00122  0.21678  0.86609
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.393866   0.043288  101.50  <2e-16 ***
## sqft         0.036044   0.001486   24.26  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.34 on 498 degrees of freedom
## Multiple R-squared:  0.5417, Adjusted R-squared:  0.5408
## F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
mean_price <- mean(data$price)
beta2 <- coef(model)["sqft"]

slope_mean <- beta2 * mean_price
mean_sqft <- mean(data$sqft)
elasticity_mean <- beta2 * mean_sqft
cat("Slope at means (a) = ", slope_mean, "\n")
```

```
## Slope at means (a) = 9.019661
```

```
cat("Elasticity at means (a) = ", elasticity_mean, "\n\n")
```

```
## Elasticity at means (a) = 0.9833701
```

The estimated log-linear model is:

$$\ln(\text{PRICE}) = 4.3939 + 0.03604 \cdot \text{SQFT} + e$$

The intercept (4.3939) represents the predicted log price when SQFT equals zero, which has no practical economic meaning.

The slope coefficient (0.03604) implies that a one-unit increase in SQFT increases house price by approximately 3.604%, holding other factors constant.

The marginal effect of SQFT on PRICE is given by:

$$\frac{\partial PRICE}{\partial SQFT} = \beta_2 \cdot PRICE$$

At the sample mean, the slope is:

$$9.019661$$

This means that at the average house price, increasing SQFT by one unit increases the price by approximately 9.02 dollars.

The elasticity of PRICE with respect to SQFT is:

$$Elasticity = \beta_2 \cdot SQFT$$

At the sample mean, the elasticity is:

$$0.9833701$$

This implies that at the average house size, a 1% increase in SQFT leads to approximately a 0.983% increase in house price.

The marginal effect is not constant and depends on the level of PRICE, which is a key feature of the log-linear model.

(b)

```
data <- data %>%
  mutate(
    ln_price = log(price),
    ln_sqft = log(sqft)
  )

model_b <- lm(ln_price ~ ln_sqft, data = data)

summary(model_b)
```

```
##
## Call:
## lm(formula = ln_price ~ ln_sqft, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.36864 -0.20849  0.00487  0.23204  1.21099
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.04965    0.15797   12.97  <2e-16 ***
## ln_sqft      1.02483    0.04839   21.18  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3643 on 498 degrees of freedom
## Multiple R-squared:  0.4738, Adjusted R-squared:  0.4728
## F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
alpha2 <- coef(model_b)["ln_sqft"]

mean_price <- mean(data$price)
mean_sqft <- mean(data$sqft)

slope_b <- alpha2 * (mean_price / mean_sqft)

elas_b <- alpha2

cat("Slope at means (b) =", slope_b, "\n")
```

```
## Slope at means (b) = 9.399923
```

```
cat("Elasticity (b) =", elas_b, "\n")
```

```
## Elasticity (b) = 1.024828
```

The estimated log-log model is:

$$\ln(PRICE) = 2.0497 + 1.0248 \cdot \ln(SQFT) + e$$

The intercept (2.0497) represents the predicted log price when $\ln(SQFT)$ equals zero, which has no clear economic interpretation.

The slope coefficient (1.0248) is the elasticity of PRICE with respect to SQFT. It implies that a 1% increase in SQFT leads to approximately a 1.0248% increase in house price, holding other factors constant.

The marginal effect of SQFT on PRICE is given by:

$$\frac{\partial PRICE}{\partial SQFT} = \alpha_2 \cdot \frac{PRICE}{SQFT}$$

At the sample mean, the slope is:

9.399923

This means that at the average values of PRICE and SQFT, increasing SQFT by one unit increases the price by approximately 9.40 dollars.

The elasticity is:

1.024828

This confirms that house price is approximately unit elastic with respect to SQFT.

Unlike the log-linear model, the elasticity in the log-log model is constant and does not depend on the level of SQFT or PRICE.

(c)

```
model_lin <- lm(price ~ sqft, data = data)

model_loglin <- lm(log(price) ~ sqft, data = data)

model_loglog <- lm(log(price) ~ log(sqft), data = data)

R2_linear <- summary(model_lin)$r.squared

yhat_loglin <- exp(fitted(model_loglin))
R2_loglin_gen <- cor(data$price, yhat_loglin)^2

yhat_loglog <- exp(fitted(model_loglog))
R2_loglog_gen <- cor(data$price, yhat_loglog)^2

cat("R² (Linear model) =", R2_linear, "\n")
```

```
## R² (Linear model) = 0.6413167
```

```
cat("Generalized R² (log-linear) =", R2_loglin_gen, "\n")
```

```
## Generalized R² (log-linear) = 0.6621612
```

```
cat("Generalized R² (log-log) =", R2_loglog_gen, "\n")
```

```
## Generalized R² (log-log) = 0.6445084
```

The R^2 from the linear model is:

0.6413167

The generalized R^2 values for the transformed models are:

- Log-linear model:

0.6621612

- Log-log model:

0.6445084

Since the dependent variables differ across models (PRICE vs. $\ln(\text{PRICE})$), the standard R^2 values are not directly comparable. Therefore, we use the generalized R^2 , defined as the squared correlation between the actual PRICE and the predicted PRICE.

Comparing the models, the log-linear model has the highest generalized R^2 , indicating that it provides the best fit among the three models.

The log-log model performs slightly better than the linear model, but not as well as the log-linear model.

Overall, the results suggest that transforming the dependent variable improves model fit, especially in the log-linear specification.

(d)

```
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

```
model_a <- lm(log(price) ~ sqft, data = data)
model_b <- lm(log(price) ~ log(sqft), data = data)
model_c <- lm(price ~ sqft, data = data)

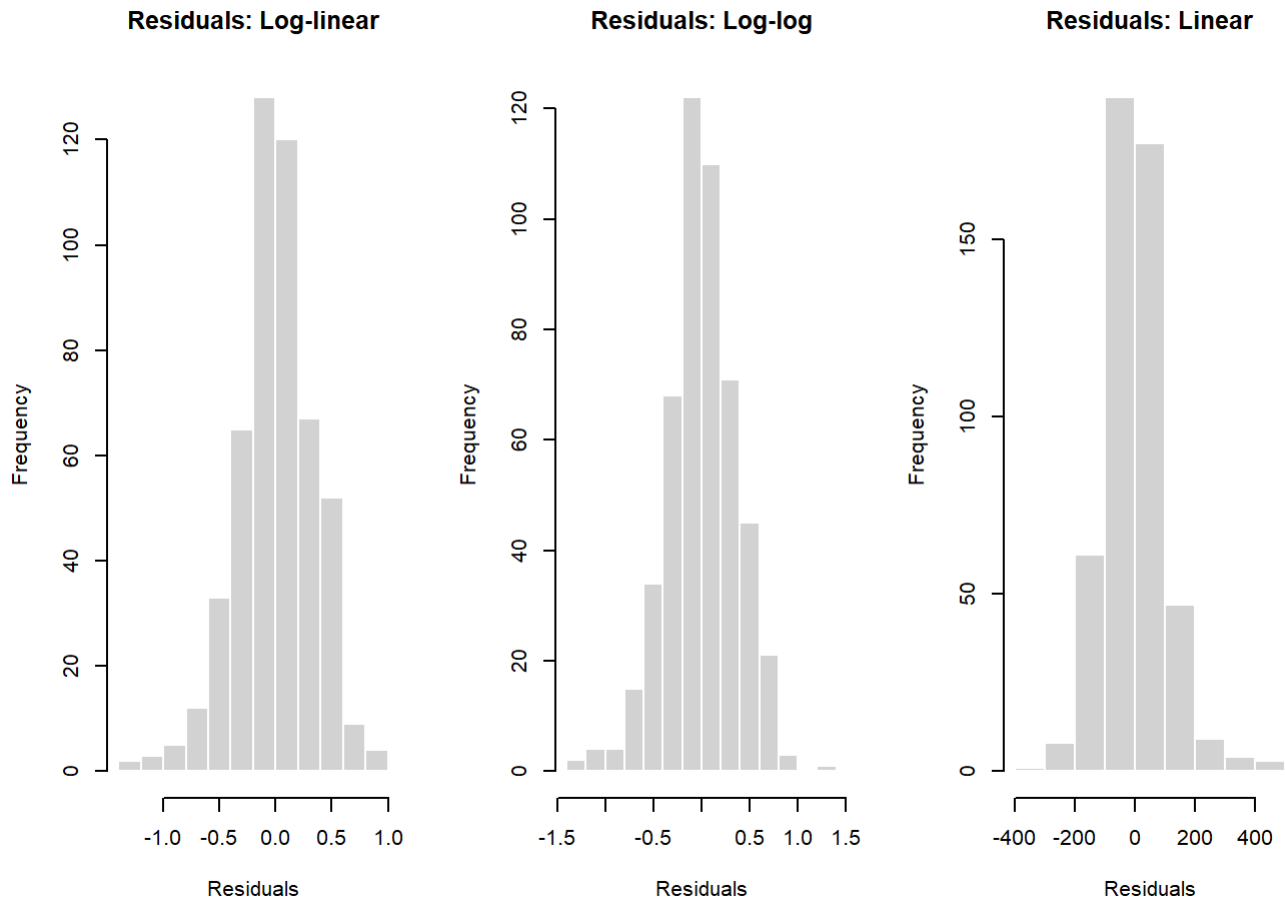
res_a <- resid(model_a)
res_b <- resid(model_b)
res_c <- resid(model_c)

par(mfrow = c(1, 3))

hist(res_a,
     main = "Residuals: Log-linear",
     xlab = "Residuals",
     col = "lightgray",
     border = "white")

hist(res_b,
     main = "Residuals: Log-log",
     xlab = "Residuals",
     col = "lightgray",
     border = "white")

hist(res_c,
     main = "Residuals: Linear",
     xlab = "Residuals",
     col = "lightgray",
     border = "white")
```



```
par(mfrow = c(1, 1))

jb_a <- jarque.bera.test(res_a)
jb_b <- jarque.bera.test(res_b)
jb_c <- jarque.bera.test(res_c)

cat("JB statistic (a) =", jb_a$statistic, "\n")
```

```
## JB statistic (a) = 26.67899
```

```
cat("p-value (a) =", jb_a$p.value, "\n\n")
```

```
## p-value (a) = 1.609651e-06
```

```
cat("JB statistic (b) =", jb_b$statistic, "\n")
```

```
## JB statistic (b) = 14.27871
```

```
cat("p-value (b) =", jb_b$p.value, "\n\n")
```

```
## p-value (b) = 0.0007932632
```

```
cat("JB statistic (c) =", jb_c$statistic, "\n")
```

```
## JB statistic (c) = 221.1115
```

```
cat("p-value (c) =", jb_c$p.value, "\n\n")
```

```
## p-value (c) = 0
```

Histograms of the least squares residuals were constructed for the three models, and Jarque–Bera tests were performed to assess normality.

The Jarque–Bera results are:

- Model (a), log-linear:
JB statistic = 26.67989, p-value = 1.609651e-06
- Model (b), log-log:
JB statistic = 14.27871, p-value = 0.0007932632
- Model (c), linear:
JB statistic = 221.1115, p-value = 0

For all three models, the p-values are far below 0.05, so the null hypothesis of normality is rejected in every case. Therefore, the residuals are not compatible with the assumption of normality.

However, the residuals from the log-log model appear to be closer to normality than those from the log-linear model, since its Jarque–Bera statistic is smaller. The linear model performs the worst, with a much larger Jarque–Bera statistic, indicating the strongest departure from normality.

Overall, although none of the three residual distributions can be considered normal, the transformed models, especially the log-log model, appear to be closer to normality than the linear model.

(e)

```
model_a <- lm(log(price) ~ sqft, data = data)
model_b <- lm(log(price) ~ log(sqft), data = data)
model_c <- lm(price ~ sqft, data = data)

res_a <- resid(model_a)
res_b <- resid(model_b)
res_c <- resid(model_c)

par(mfrow = c(1, 3))

plot(data$sqft, res_a,
     main = "Residuals vs SQFT (log-linear)",
     xlab = "SQFT",
     ylab = "Residuals",
     pch = 16, col = "gray")

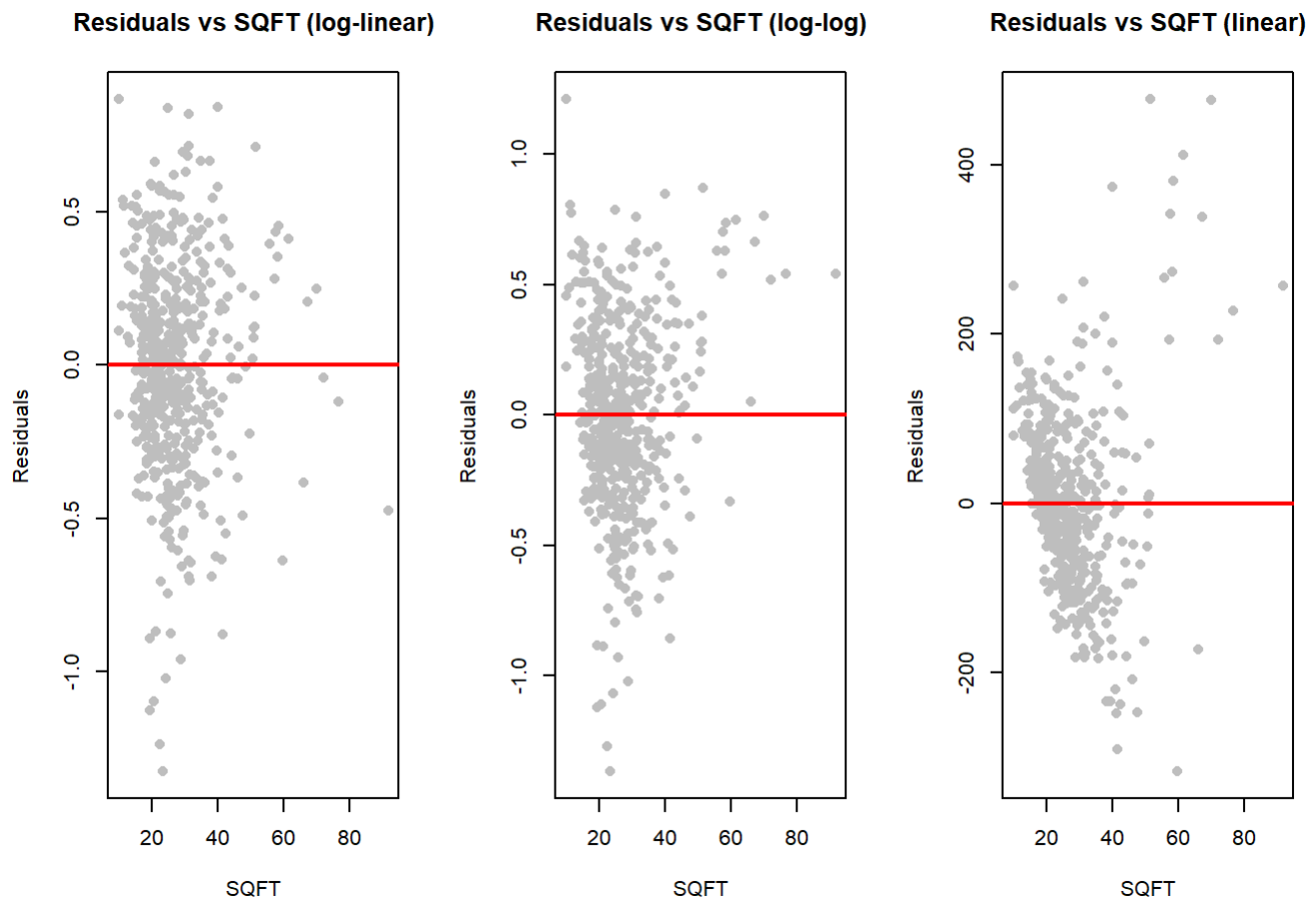
abline(h = 0, col = "red", lwd = 2)

plot(data$sqft, res_b,
     main = "Residuals vs SQFT (log-log)",
     xlab = "SQFT",
     ylab = "Residuals",
     pch = 16, col = "gray")

abline(h = 0, col = "red", lwd = 2)

plot(data$sqft, res_c,
     main = "Residuals vs SQFT (linear)",
     xlab = "SQFT",
     ylab = "Residuals",
     pch = 16, col = "gray")

abline(h = 0, col = "red", lwd = 2)
```

```
par(mfrow = c(1, 1))
```

The residuals from each model are plotted against SQFT to assess potential patterns.

For the **linear model**, the residuals display a clear fan-shaped pattern, with the spread of residuals increasing as SQFT increases. This indicates the presence of heteroskedasticity, meaning the variance of the errors is not constant.

For the **log-linear model**, the residuals appear more evenly distributed around zero, with a more stable spread across different values of SQFT. However, there is still some mild dispersion as SQFT increases.

For the **log-log model**, the residuals show the most stable variance among the three models. The spread is relatively uniform, and there is no strong visible pattern, suggesting that heteroskedasticity is less severe.

Overall, the linear model exhibits the strongest heteroskedasticity, while the log transformations (especially the log-log model) improve the distribution of residuals and provide a more appropriate specification.

(f)

```
new_data <- data.frame(sqft = 27)

pred_c <- predict(model_c, newdata = new_data)

cat("--- Linear Model ---\n")
```

```
## --- Linear Model ---
```

```
cat("Predicted PRICE =", pred_c, "\n\n")
```

```
## Predicted PRICE = 246.4557
```

```
pred_log_a <- predict(model_a, newdata = new_data)
```

```
sigma2_a <- summary(model_a)$sigma^2
```

```
pred_a <- exp(pred_log_a + sigma2_a / 2)
```

```
cat("--- Log-Linear Model ---\n")
```

```
## --- Log-Linear Model ---
```

```
cat("Predicted PRICE =", pred_a, "\n\n")
```

```
## Predicted PRICE = 226.9817
```

```
pred_log_b <- predict(model_b, newdata = new_data)
```

```
sigma2_b <- summary(model_b)$sigma^2
```

```
pred_b <- exp(pred_log_b + sigma2_b / 2)
```

```
cat("--- Log-Log Model ---\n")
```

```
## --- Log-Log Model ---
```

```
cat("Predicted PRICE =", pred_b, "\n")
```

```
## Predicted PRICE = 243.1514
```

(g)

```
new_data <- data.frame(sqft = 27)
```

```
mod_a <- lm(log(price) ~ sqft, data = data)
```

```
mod_b <- lm(log(price) ~ log(sqft), data = data)
```

```
mod_c <- lm(price ~ sqft, data = data)
```

```
pi_c <- predict(mod_c, newdata = new_data, interval = "prediction", level = 0.95)
```

```
cat("--- Linear Model ---\n")
```

```
## --- Linear Model ---
```

```
print(pi_c)
```

```
##          fit      lwr      upr
## 1 246.4557 44.27727 448.6341
```

```
cat("\n")
```

```
pi_log_a <- predict(mod_a, newdata = new_data, interval = "prediction", level = 0.95)

sigma2_a <- summary(mod_a)$sigma^2
point_a <- exp(pi_log_a[1, "fit"] + sigma2_a / 2)

lwr_a <- exp(pi_log_a[1, "lwr"])
upr_a <- exp(pi_log_a[1, "upr"])

cat("--- Log-Linear Model ---\n")
```

```
## --- Log-Linear Model ---
```

```
cat("Point prediction =", point_a, "\n")
```

```
## Point prediction = 226.9817
```

```
cat("95% prediction interval = [", lwr_a, ",", upr_a, "]\n\n")
```

```
## 95% prediction interval = [ 109.7685 , 418.1167 ]
```

```
pi_log_b <- predict(mod_b, newdata = new_data, interval = "prediction", level = 0.95)

sigma2_b <- summary(mod_b)$sigma^2
point_b <- exp(pi_log_b[1, "fit"] + sigma2_b / 2)

lwr_b <- exp(pi_log_b[1, "lwr"])
upr_b <- exp(pi_log_b[1, "upr"])

cat("--- Log-Log Model ---\n")
```

```
## --- Log-Log Model ---
```

```
cat("Point prediction =", point_b, "\n")
```

```
## Point prediction = 243.1514
```

```
cat("95% prediction interval = [", lwr_b, ",", upr_b, "]\n\n")
```

```
## 95% prediction interval = [ 111.1406 , 465.8406 ]
```

(h)

Based on the results in this problem, the **log-linear** and **log-log** models both perform better than the simple linear model.

First, the transformed models fit the data slightly better when we compare the generalized R^2 values. The generalized R^2 is highest for the **log-linear model** (0.6622), followed by the **log-log model** (0.6445), while the linear model has the lowest fit (0.6413).

Second, the residual analysis suggests that the **linear model is the least appropriate**. Its residual plot against SQFT shows a clear fan-shaped pattern, indicating substantial heteroskedasticity. The Jarque–Bera statistic is also much larger for the linear model (221.1115), showing the strongest departure from normality.

Both transformed models improve the residual behavior. The **log-log model** has the smallest Jarque–Bera statistic (14.2787), so its residuals are closest to normality. It also shows the most stable residual variance across values of SQFT. However, the **log-linear model** has the best overall fit according to the generalized R^2 .

If I had to choose one specification, I would use the **log-linear model**. It provides the best fit among the three models and also improves the heteroskedasticity problem relative to the linear model. Although the log-log model has residuals that are somewhat closer to normality, the log-linear model performs slightly better overall in explaining house price.

In addition, the log-linear model is easy to interpret: a one-unit increase in SQFT changes house price by an approximate percentage amount. For these data, that specification seems to provide the best balance of fit, interpretation, and residual behavior.